B7 Exploratory analysis done well

Profiling, visual checks and documented assumptions.

B7-L1 Profile a dataset before you analyse it

Shape and grain, dtypes, missingness, cardinality and ranges, in code, before any analysis.

```python
def profile(df: pd.DataFrame) -> pd.DataFrame:
    return pd.DataFrame({
        "dtype": df.dtypes.astype(str),
        "missing_share": df.isna().mean(),
        "n_unique": df.nunique(),
    })
```

Five profiling questions	Shape and grain, dtypes, missingness, cardinality, ranges.
nunique() and missing values	Excluded by default. dropna=False counts NaN as a value.
Count every row in a duplicate group	df.duplicated(subset=key, keep=False).sum().
Why save the profile?	Compare it with the next delivery to spot source changes early.

B7-L2 Plot before you summarise

Plot distributions, pairs and time series before summarising, with functions that return labelled figures.

Anscombe's quartet	Same mean, variance and correlation (0.816), different shapes. Plot before summarising.
Four first plots	Histograms, scatter plots of modelled pairs, lines over waves, missingness bars.
Testable plotting code	fig, ax = plt.subplots(), set labels on ax, return fig. No plt.show() inside.
Histogram bins	Choose on purpose and try several. Too few hide spikes at codes.

B7-L3 Write assumptions down, then turn them into checks

Log each assumption with evidence and owner, implement its handling per column, and check it on every run.

```yaml
- id: A3
  statement: "score uses 1 to 10; 98 = refused, 99 = don't know"
  evidence: "Questionnaire v4, Q12; profile of waves 1 to 11 shows only 1-10, 98, 99"
  handling: "98 and 99 become missing before any statistic"
  check: "score in 1..10 or missing, after recoding"
  owner: "analysis team"
  last_verified: 2026-09-01
```

```python
MISSING_CODES: dict[str, list[int]] = {"score": [98, 99]}   # from config, per A3


def recode_missing(df: pd.DataFrame, codes: dict[str, list[int]]) -> pd.DataFrame:
    return df.assign(**{col: df[col].mask(df[col].isin(vals)) for col, vals in codes.items()})
```

Fields of an assumptions log entry	id, statement, evidence, handling, check, owner, last_verified.

Recode missing codes safely	Per column: df[col].mask(df[col].isin(codes)). Never a whole-frame replace.
Three kinds of assumption	Structure (grain, keys), coding (codes, scales, units), scope (who is included).
From assumption to code	Handling in config-driven code, and a QA check that runs on every delivery.